

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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MODEL PRACTICAL EXAMINATION – SET 6

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application for Online Super Market using any Cloud Service Provider to demonstrate SaaS, with the template of Customer Name, Address, Phone, Email, Items, Quantity, etc. along with the item classification form.

AIM:

To create a simple cloud-based Online Super Market application using Zoho Creator to demonstrate the Software as a Service (SaaS) model.

PROCEDURE

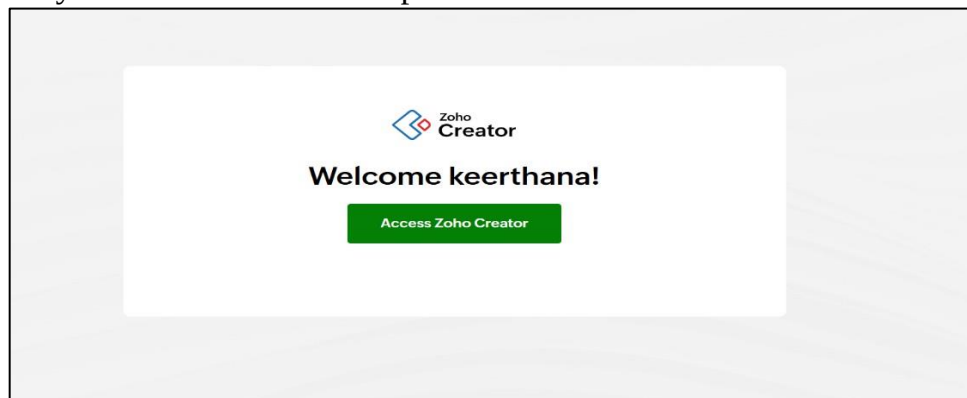
1. Login to Zoho Creator and create a new application.
2. Create Customer Details, Item Classification, and Order Details forms.
3. Add the required fields and enter sample records.
4. Create Reports and Dashboard for the application.
5. Test the application and verify the output.

Implemented

STEP 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 2: Create a New Application

Create a new application and enter Online Super Market Management System as the application name.

The screenshot shows the "Create from scratch" form in Zoho Creator. At the top, there are two radio buttons, with the first one selected. Below this, the text "Create from scratch" is displayed. The form has a section for "Application Name" with a red asterisk indicating it is required. The text "Super market management system" is entered in the input field. Below the input field, there is a checkbox labeled "Enable Environment" with a link "Learn More" next to it. At the bottom of the form, there is a blue button labeled "Create Application".

STEP 3: Create Customer Details Form

Create the Customer Details form and add the required customer fields.

The screenshot displays the 'Customer Details' form within the 'Super Market Management System'. The left sidebar contains a navigation menu with options: Dashboard, Order Details, Sales Reports, Customer Details (expanded), All Customer Details, Item Classification, Items, Sales Orders, and Customers. The 'Customer Details' section is active, showing a form with the following fields: Customer Name, Address (Address Line 1), Phone Number (with a dropdown for country code, currently showing +91 and the number 81234 56789), Email, and Linked Contact (a dropdown menu with '-Select-' as the current selection). At the bottom of the form are 'Submit' and 'Reset' buttons. The top of the browser window shows the system name and a trial expiration notice: 'Trial expires in 15 days'.

STEP 4: Create Item Classification Form

Create the Item Classification form with item name, category, price, and available quantity.

The screenshot displays the 'Item Classification' form within the 'Super Market Management System'. The left sidebar is identical to the previous screenshot, with 'Item Classification' now selected. The 'Item Classification' section is active, showing a form with the following fields: Brand, Category, Cost Price, Created Time, Dimension Unit, EAN, Height, HSN Or SAC Code, ISBN, Last Modified Time, and Length. The 'Cost Price', 'Created Time', 'EAN', 'Height', 'HSN Or SAC Code', 'ISBN', 'Last Modified Time', and 'Length' fields have placeholder text indicating required formats (e.g., 'dd-MM-yyyy HH:mm:ss' for time fields). The 'Brand' field is currently empty. At the bottom of the form are 'Submit' and 'Reset' buttons. The top of the browser window shows the system name and a trial expiration notice: 'Trial expires in 15 days'.

STEP 5: Create Order Details Form

Create the Order Details form and add customer, item, quantity, price, total amount, order date, and payment status fields.

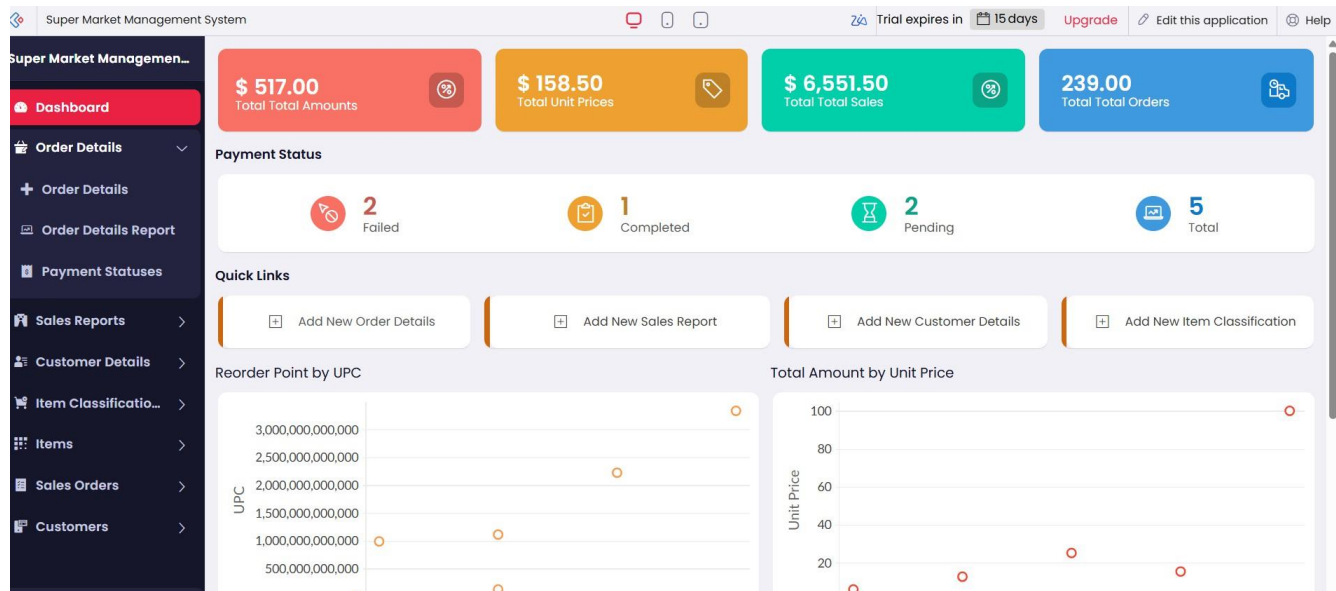
The screenshot shows the 'Order Details' form within the 'Super Market Management System'. The form includes the following fields:

- Order Date: A date input field with a placeholder 'dd-MMM-yyyy'.
- Quantity: A numeric input field with a placeholder '#####'.
- Unit Price: A numeric input field with a placeholder '#####.##'.
- Total Amount: A numeric input field with a placeholder '#####.##'.
- Payment Status: A dropdown menu with '-Select-' as the selected option.
- Customer: A dropdown menu with '-Select-' as the selected option.
- Item: A dropdown menu with '-Select-' as the selected option.
- Linked Sales Order: A dropdown menu with '-Select-' as the selected option.

At the bottom of the form, there are two buttons: 'Submit' (in red) and 'Reset' (in light gray). The left sidebar contains a navigation menu with options like Dashboard, Order Details, Order Details Report, Payment Statuses, Sales Reports, Customer Details, Item Classification, Items, Sales Orders, and Customers.

STEP 6: Create Final Dashboard

Create the Online Super Market Dashboard with summary panels and charts.



RESULT

Thus, the Online Super Market SaaS application was successfully created using Zoho Creator with three forms, reports, dashboard, and cloud-based data storage.

2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.

AIM

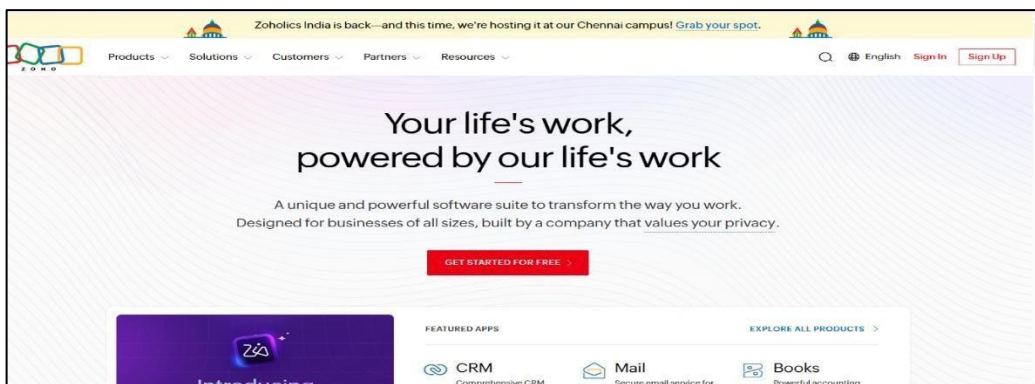
To create a cloud-based Mark Sheet Processing System using Zoho Creator to demonstrate the SaaS model.

PROCEDURE

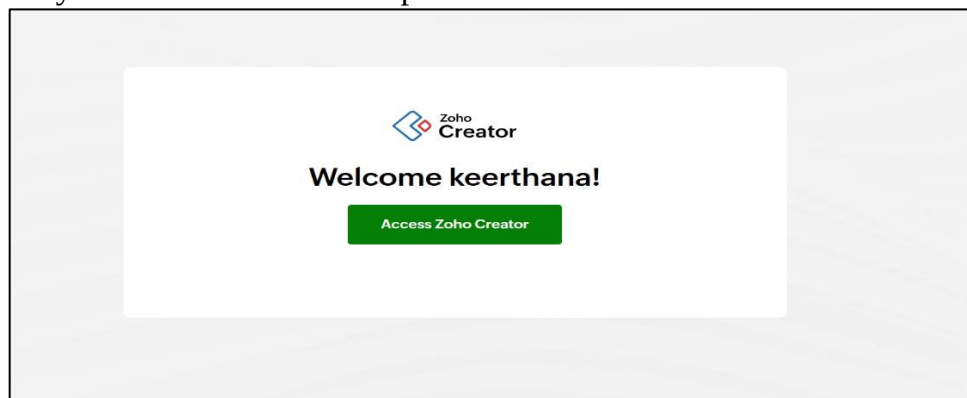
1. Login to Zoho Creator and create a new application.
2. Create Student Details, Subject Details, Mark Details, and Mark Sheet Summary forms.
3. Add the required fields and enter sample records.
4. Create Reports and Dashboard.
5. Test the application and verify the output.

STEPS

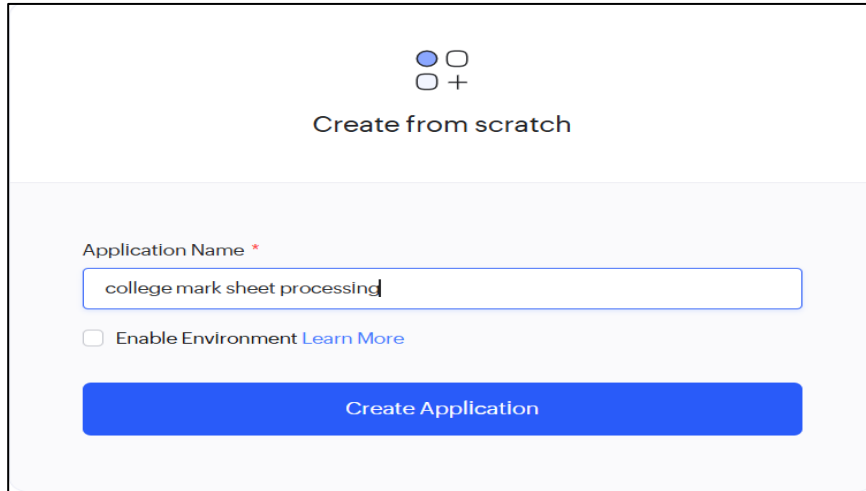
Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator

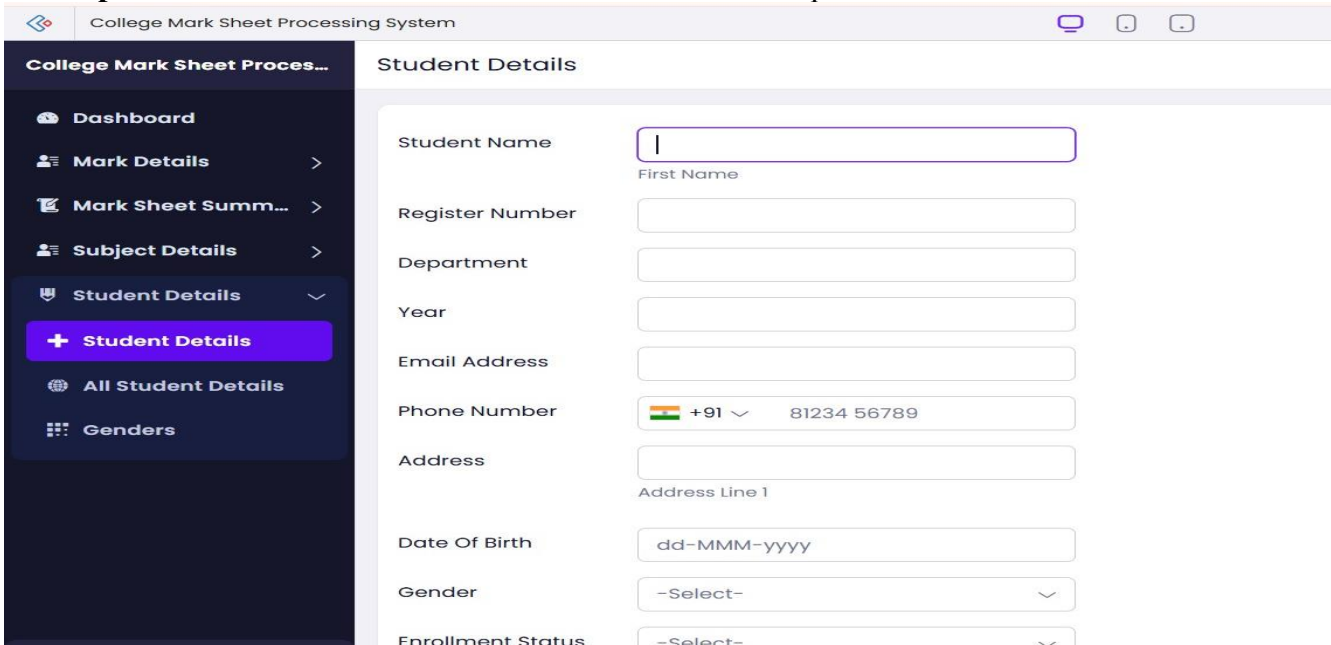


Step 3: Create a new application named Mark Sheet Processing System.



The screenshot shows a web interface for creating a new application. At the top, there is a logo consisting of two blue circles and a plus sign, with the text "Create from scratch" below it. The main form area has a label "Application Name *" followed by a text input field containing "college mark sheet processing". Below the input field is a checkbox labeled "Enable Environment" with a link "Learn More" next to it. At the bottom of the form is a large blue button labeled "Create Application".

Step 4: Create the Student Details form and add the required fields.



The screenshot shows a web application titled "College Mark Sheet Processing System". On the left is a dark sidebar with a menu containing: "Dashboard", "Mark Details", "Mark Sheet Summ...", "Subject Details", "Student Details" (expanded), "+ Student Details" (highlighted), "All Student Details", and "Genders". The main content area is titled "Student Details" and contains a form with the following fields: "Student Name" (with a sub-label "First Name"), "Register Number", "Department", "Year", "Email Address", "Phone Number" (with a dropdown for country code set to "+91" and a value "81234 56789"), "Address" (with a sub-label "Address Line 1"), "Date Of Birth" (with a placeholder "dd-MMM-yyyy"), "Gender" (with a dropdown "-Select-"), and "Enrollment Status" (with a dropdown "-Select-").

Step 5: Create the **Subject Details** form and add the required fields.

College Mark Sheet Processing System

College Mark Sheet Proces...

Subject Details

Dashboard

Mark Details >

Mark Sheet Summ... >

Subject Details >

+ Subject Details

All Subject Details

Subject Types

Student Details >

Subject Code

Subject Name

Maximum Marks

Semester

Subject Type

Credits

Instructor Name

First Name

Description

Start Date

#####

-Select-

#####

dd-MMM-yyyy

Step 6: Create the **Mark Details** form and add the required fields.

College Mark Sheet Processing System

College Mark Sheet Proces...

Mark Details

Dashboard

Mark Details >

+ Mark Details

All Mark Details

Grades

Mark Sheet Summ... >

Subject Details >

Student Details >

Student

Subject

Marks Obtained

Maximum Marks

Grade

Result

Exam Date

Exam Center

Internal Marks

External Marks

Remarks

-Select-

-Select-

#####

#####

-Select-

-Select-

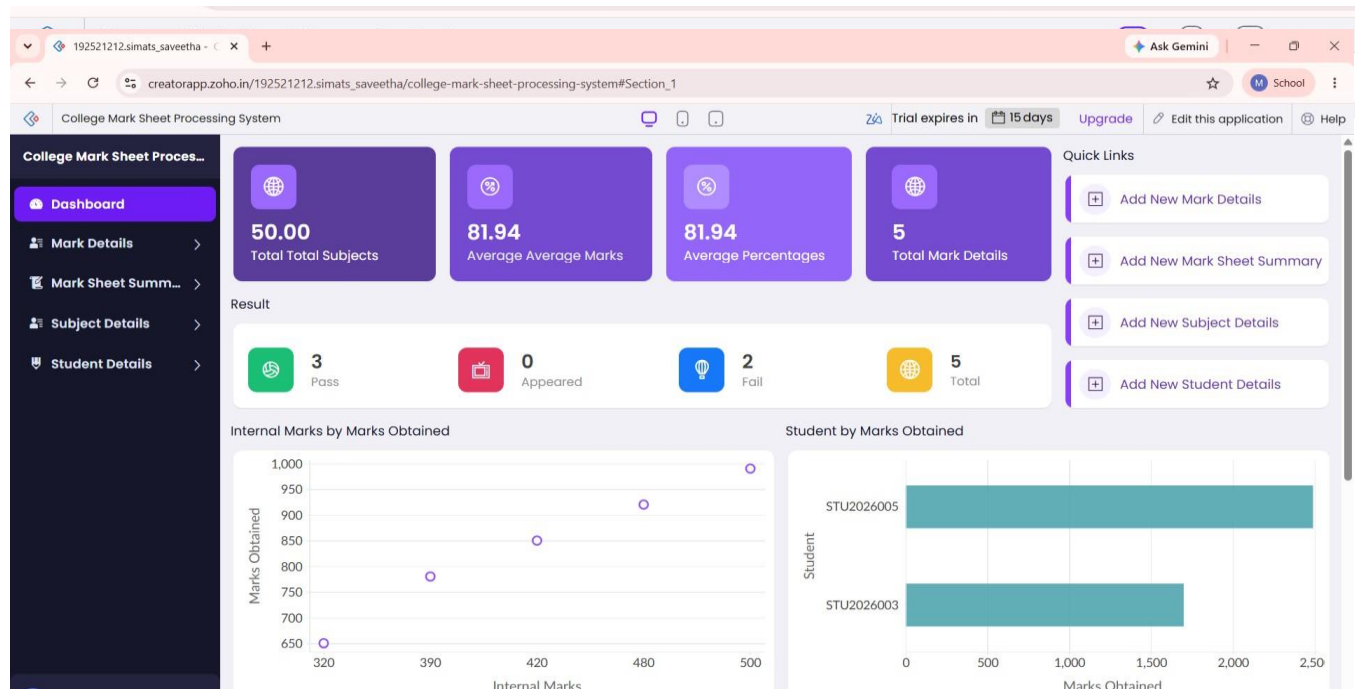
dd-MMM-yyyy

#####

#####

Step 7: Create the Mark Sheet Summary form and add the required fields.

Step 8: Create the Dashboard.



RESULT

Thus, the Mark Sheet Processing System was successfully created using Zoho Creator with four forms, reports, and dashboard.

3. Install VirtualBox with different flavours of Linux or Windows OS on top of Windows 10 using custom installation.

AIM

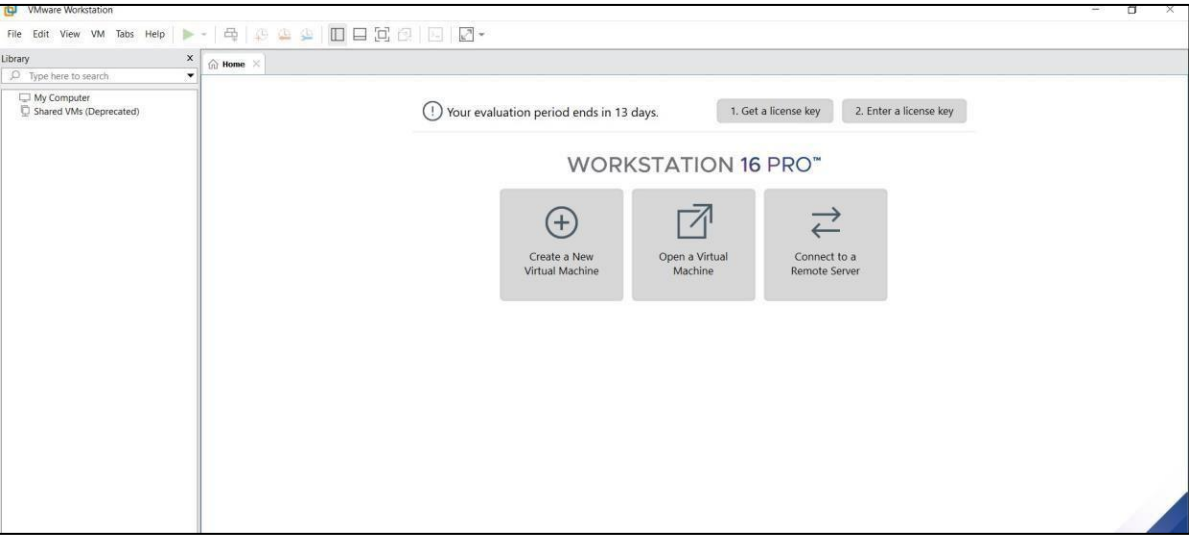
To demonstrate virtualization by installing a Type-2 Hypervisor and creating and configuring a virtual machine with a host operating system.

PROCEDURE

- 1. Download and install VMware Workstation as a Type-2 Hypervisor.
- 2. Download Ubuntu or Tiny OS as an ISO image file.
- 3. Open VMware Workstation and create a new virtual machine.
- 4. Perform the basic configuration settings.
- 5. Create the virtual machine and launch it.

STEPS

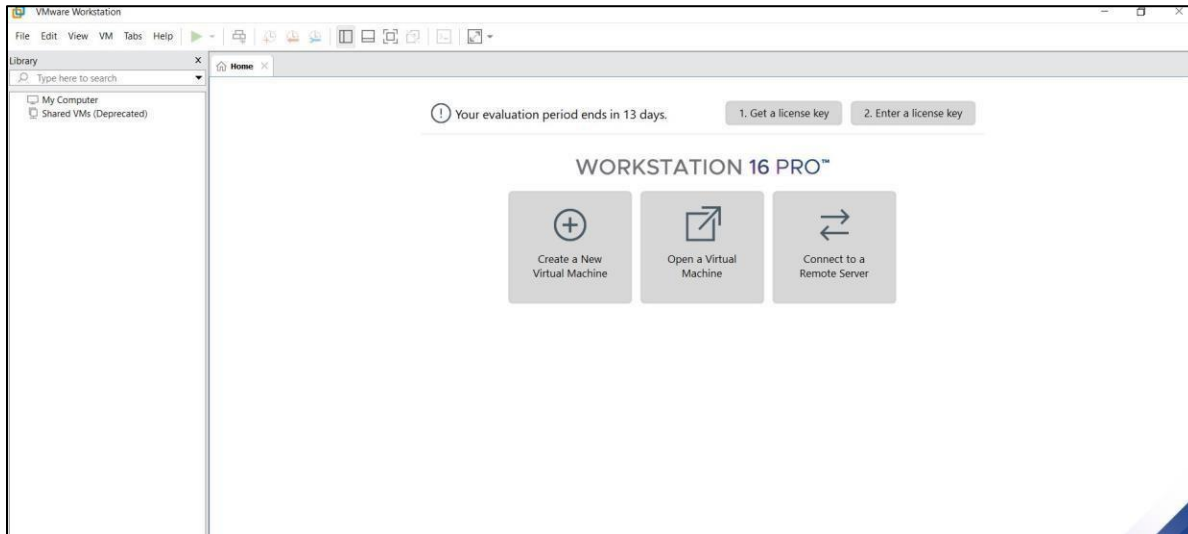
Step 1: Download and install VMware Workstation as a Type-2 Hypervisor.



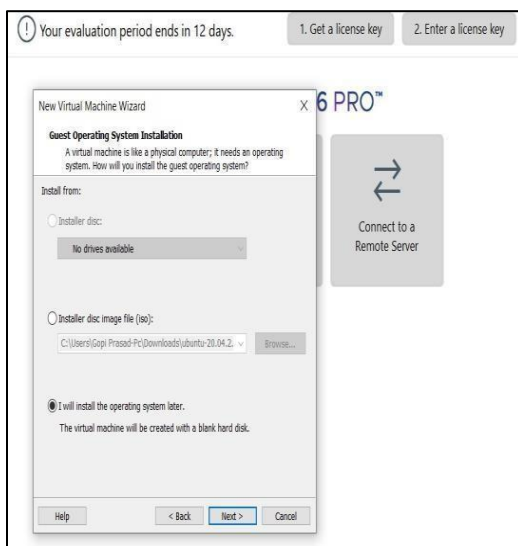
Step 2: Download Ubuntu or Tiny OS as an ISO image file.

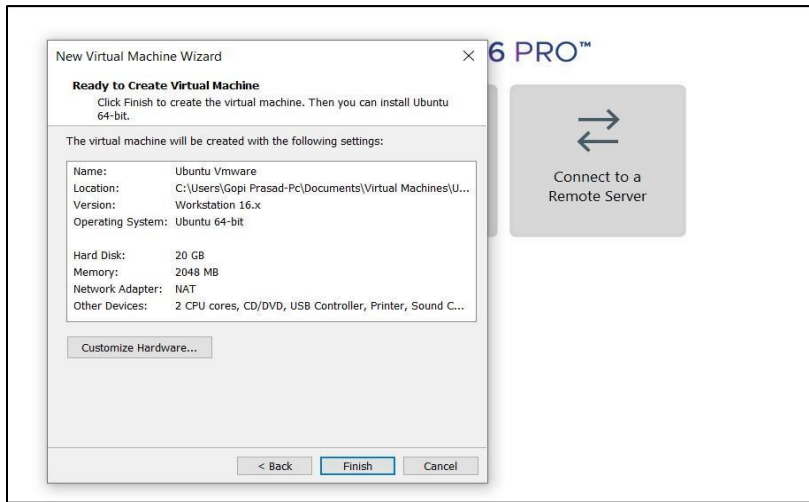
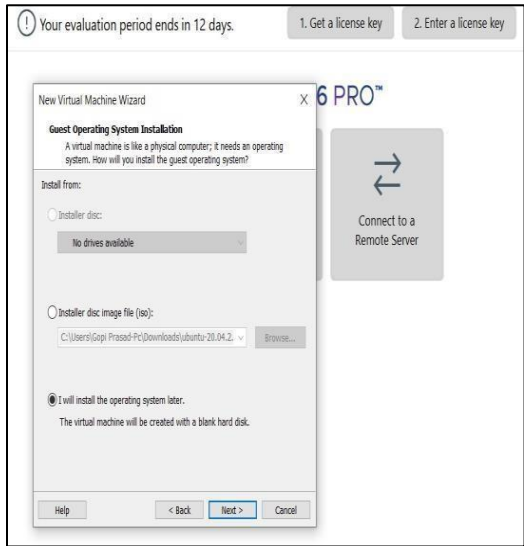
Index of /11.x/x86/release/			
..			
distribution_files/	09-Feb-2020 11:50	-	
src/	03-Dec-2019 11:14	-	
Core-11.1.iso	01-Apr-2020 07:49	14757888	
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48	
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639	
Core-current.iso	01-Apr-2020 07:49	14757888	
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656	
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52	
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358	
CorePlus-current.iso	01-Apr-2020 07:50	216006656	
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944	
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52	
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301	
TinyCore-current.iso	01-Apr-2020 07:50	19922944	

Step 3: Open VMware Workstation and select **Create New VM**.

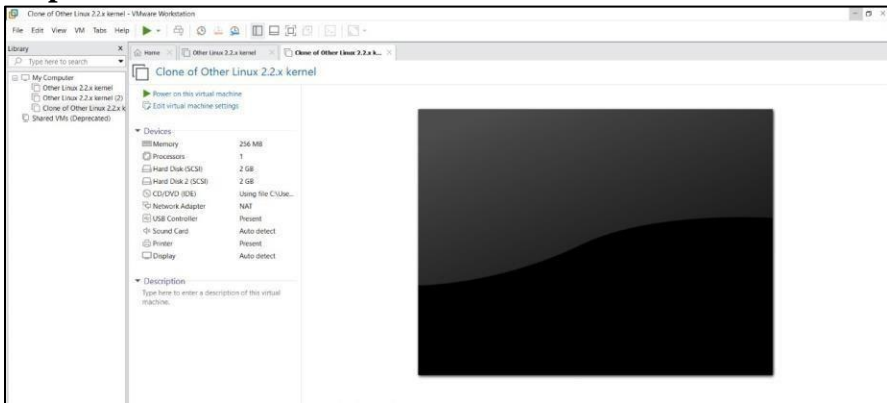


Step 4: Perform the basic configuration settings.

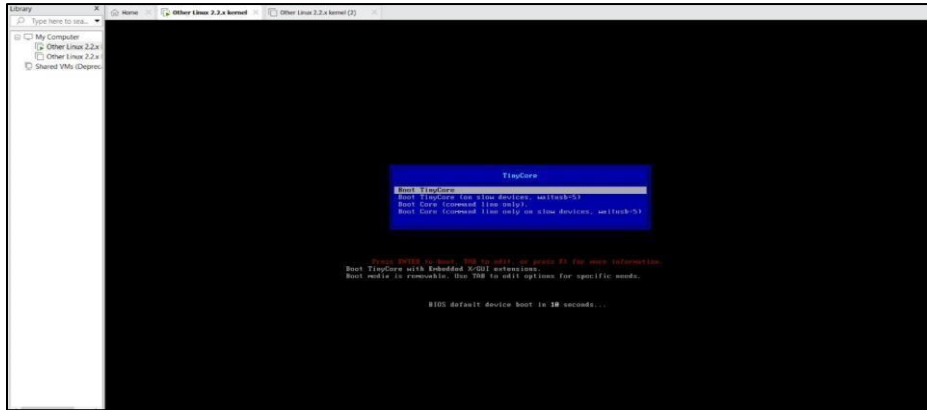




Step 5: Create the virtual machine.



Step 6: Launch the virtual machine.



RESULT

Thus, the virtual machine was successfully created and configured using VMware Workstation as a Type-2 Hypervisor.

4. Write a procedure to run virtual machines of different configurations and check how many virtual machines can be utilized at a particular time.

AIM

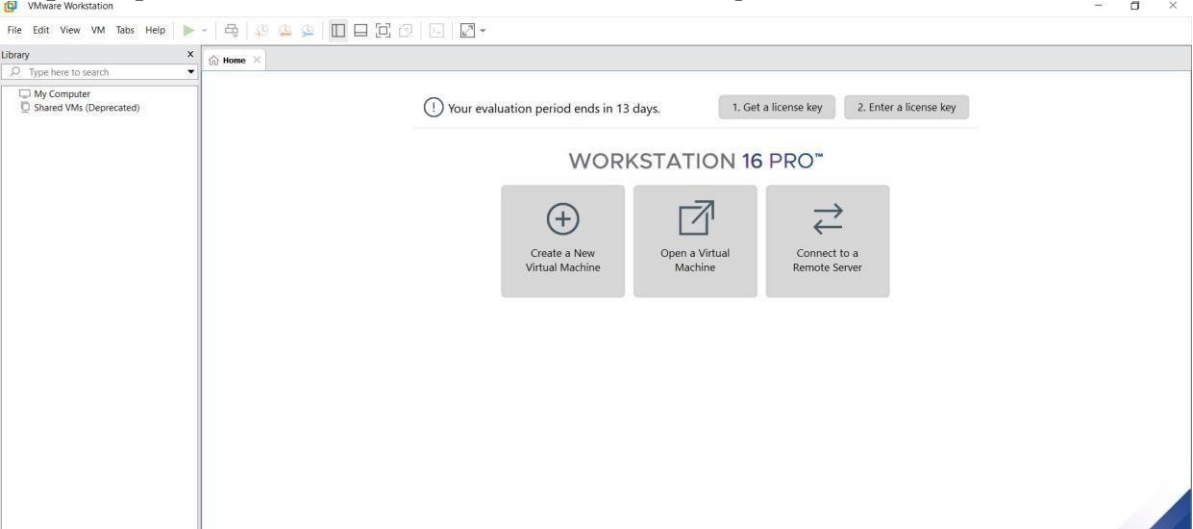
To run virtual machines with different configurations and check how many virtual machines can be utilized at a particular time.

PROCEDURE

- 1. Open VMware Workstation and create virtual machines with different configurations.
- 2. Allocate different CPU, RAM, and storage to the virtual machines.
- 3. Start the virtual machines and observe their performance.
- 4. Run multiple virtual machines simultaneously.
- 5. Check the number of virtual machines that can run at the same time.

STEPS

Step 1: Open VMware Workstation and select the required virtual machine.

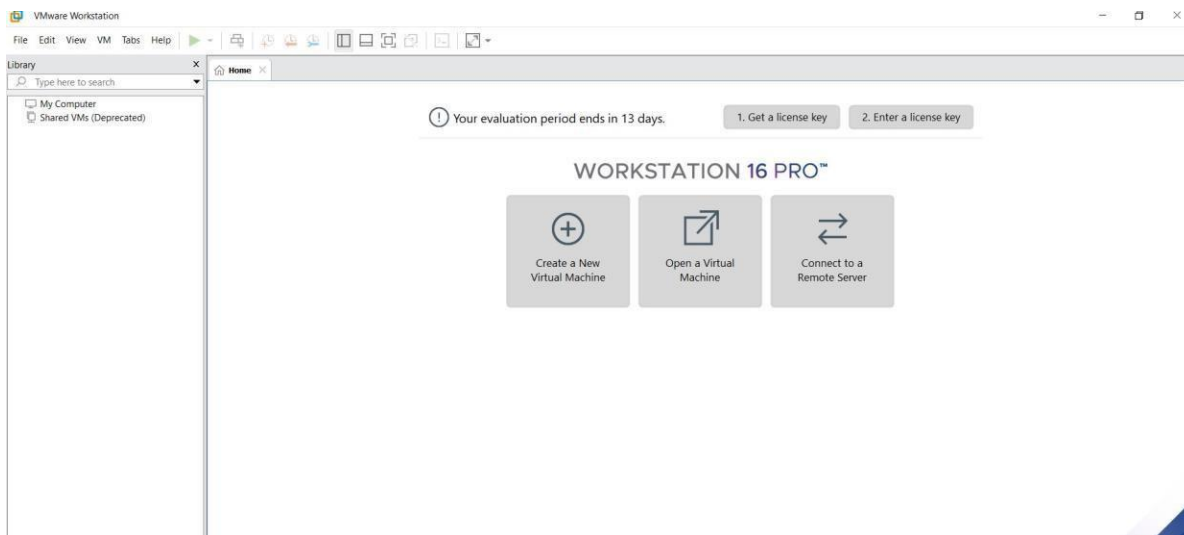


Step 2: Configure the CPU, RAM, and storage for the virtual machine.

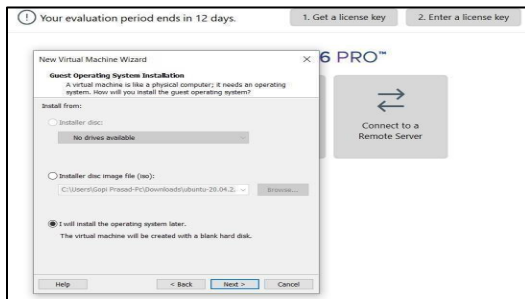
Index of /11.x/x86/release/

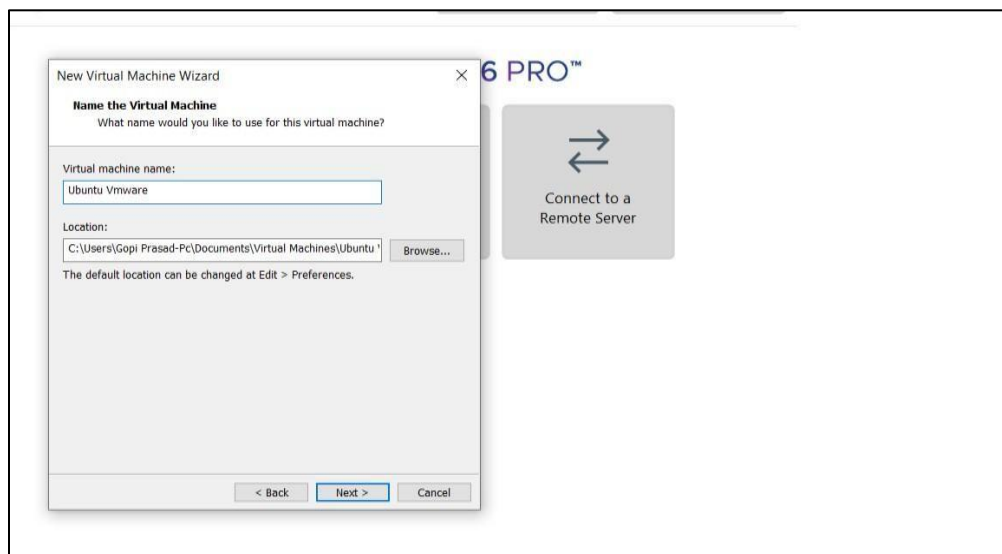
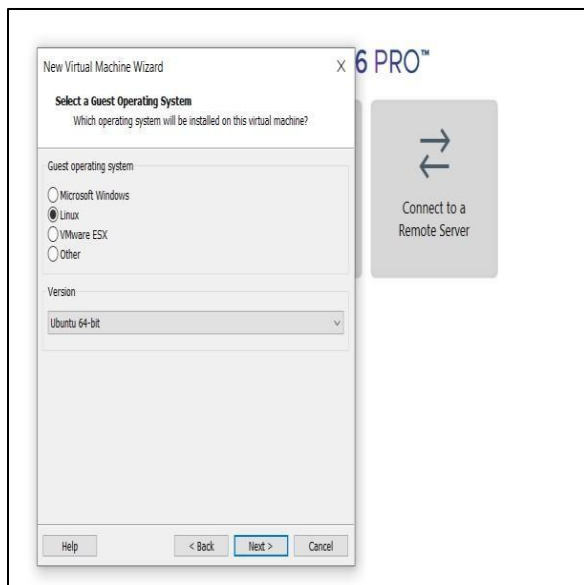
..		
distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

Step 3: Start the virtual machine.

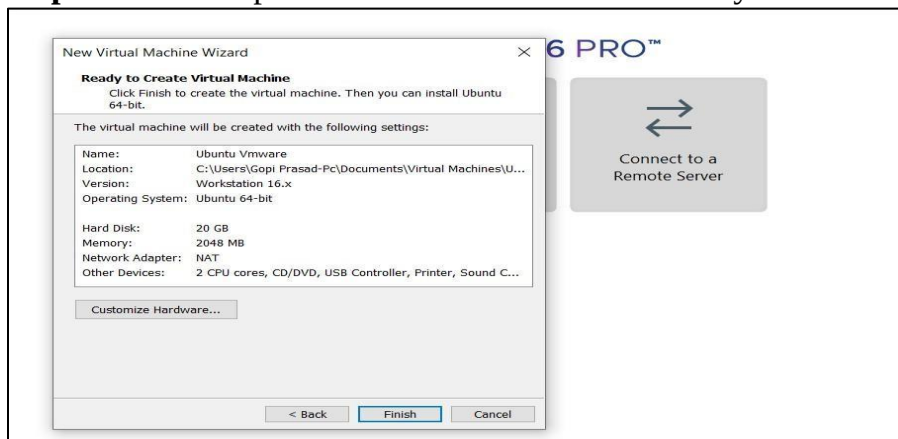


Step 4: Create or open another virtual machine with a different configuration.

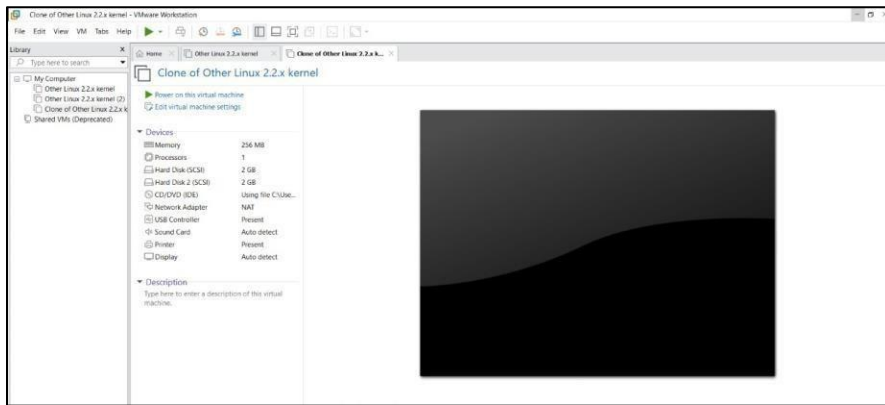




Step 5: Run multiple virtual machines simultaneously and observe the system resources.



Step 6: Check how many virtual machines can be utilized at the same time.



RESULT

Thus, virtual machines with different configurations were successfully run, and the number of virtual machines that could be utilized simultaneously was checked.

